



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

Continuous Assessment Test-I, Winter 2019-2020

CSE2003- Data Structures and Algorithms

Slot: B2

Exam Duration: 90 Minutes

Maximum Marks : 50

Answer all questions (5*10 = 50)

1. Define Big O, Big Omega and big theta notations. Discuss the same graphically.
2. a. Consider a linear linked list with n elements. Devise an algorithm to count number of nodes and find sum of all values in the list and display it. (6 Marks)
b. With neat diagrams represent all the following operations on a queue of size. Justify the output after each operation. (4 Marks)
dequeue(), enqueue(X), enqueue(B), dequeue(), enqueue(K), display(),
dequeue(), display()
3. Convert the following infix expression to postfix expression using stack. Show the stack trace in each step.

$$\mathbf{a+b*c+(d*e+f)*g}$$

4. a. Define algorithm. List and inscribe the properties of an algorithm. (5 Marks)
b. Devise linear search algorithm and inscribe the proof of correctness for the specified algorithm. (5 Marks)
5. a. Elucidate the general plan of analyzing the efficiency of recursive algorithm. Suggest a recursive algorithm to find factorial of a number and derive its efficiency. (5 Marks)
b. Solve the following recurrence relation T(n) using recursion tree method.

(5 Marks)

$$T(n)=\begin{cases} 2T(n/2) + n & , n>1 \\ 1 & , n=1 \end{cases}$$